The Residual Discontinuity Test for *p*-Hacking

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Reported precision is a choice

Table: Regression of test scores on class size

	Krueger (1999)	Replications using different computations of SE				
	(1)	(2) Bootstrap of class clusters	(3) Class clusters	(4) School clusters	(5) Huber-White SE	(6) Plain vanilla SE
Small class	4.82	4.82	4.82	4.82	4.82	4.82
	(2.19)	(2.00)	(1.88)	(1.38)	(0.79)	(0.76)
Sample	5,861	5,861	5,861	5,861	5,861	5,861

Notes: Dependent variable: test score percentile. Standard errors (SE) in parentheses.

Meta-analysis of the class size effect (JOLE)

Available at meta-analysis.cz/class. Spoiler: effect size tiny.

Changing controls changes precision

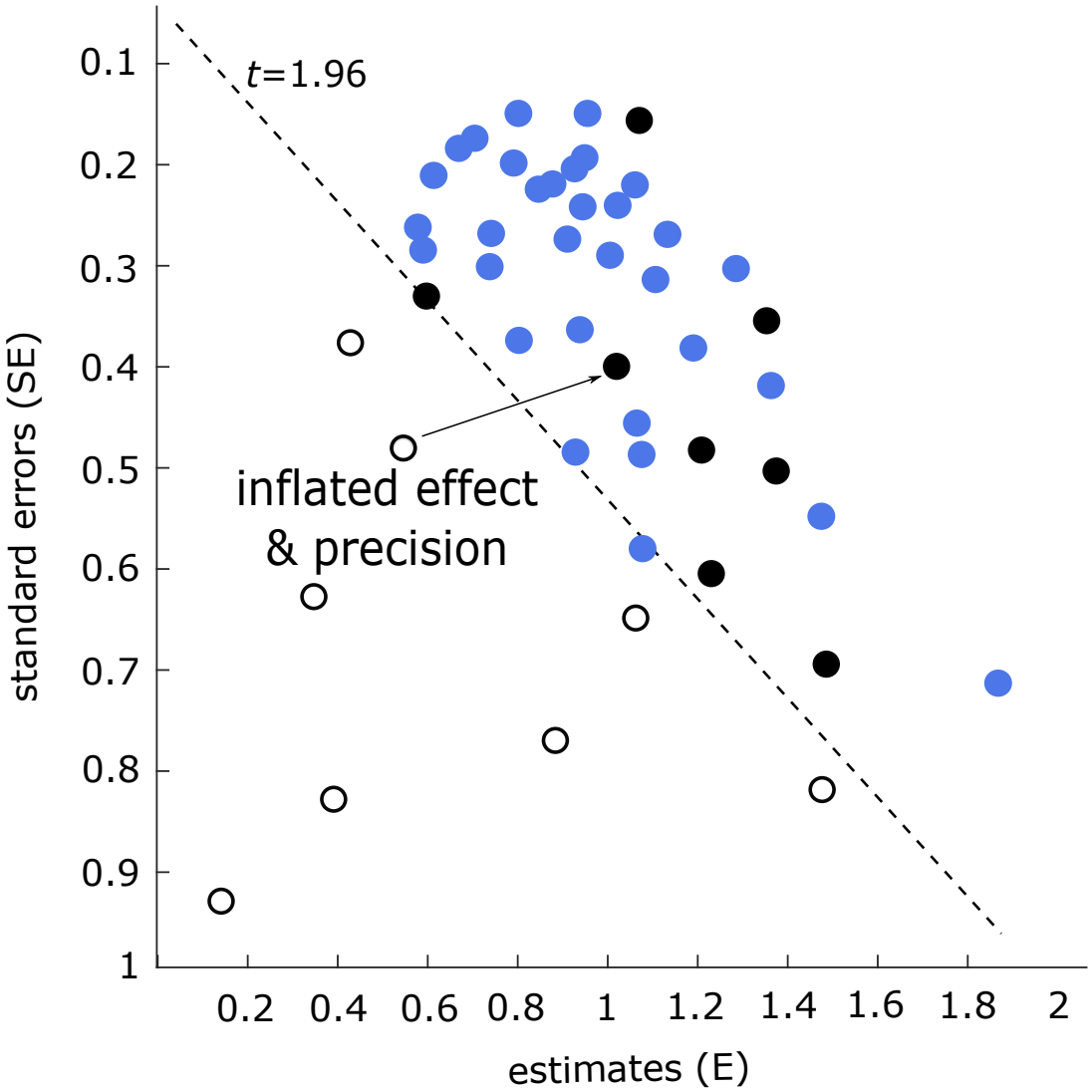
Table: Regression of test scores on class size

	(1)	(2)	(3)	(4)
Small class (treatment)	4.82	5.37	5.36	5.37
	(2.19)	(1.26)	(1.21)	(1.19)
White/Asian			8.35	8.44
			(1.35)	(1.36)
Girl			4.48	4.39
			(0.63)	(0.63)
Free lunch			-13.15	-13.07
			(0.77)	(0.77)
White teacher				-0.57
				(2.1)
Teacher experience				0.26
				(0.10)
Master's degree				-0.51
				(1.06)
School intercepts	No	Yes	Yes	Yes
Sample	5,861	5,861	5,861	5,861

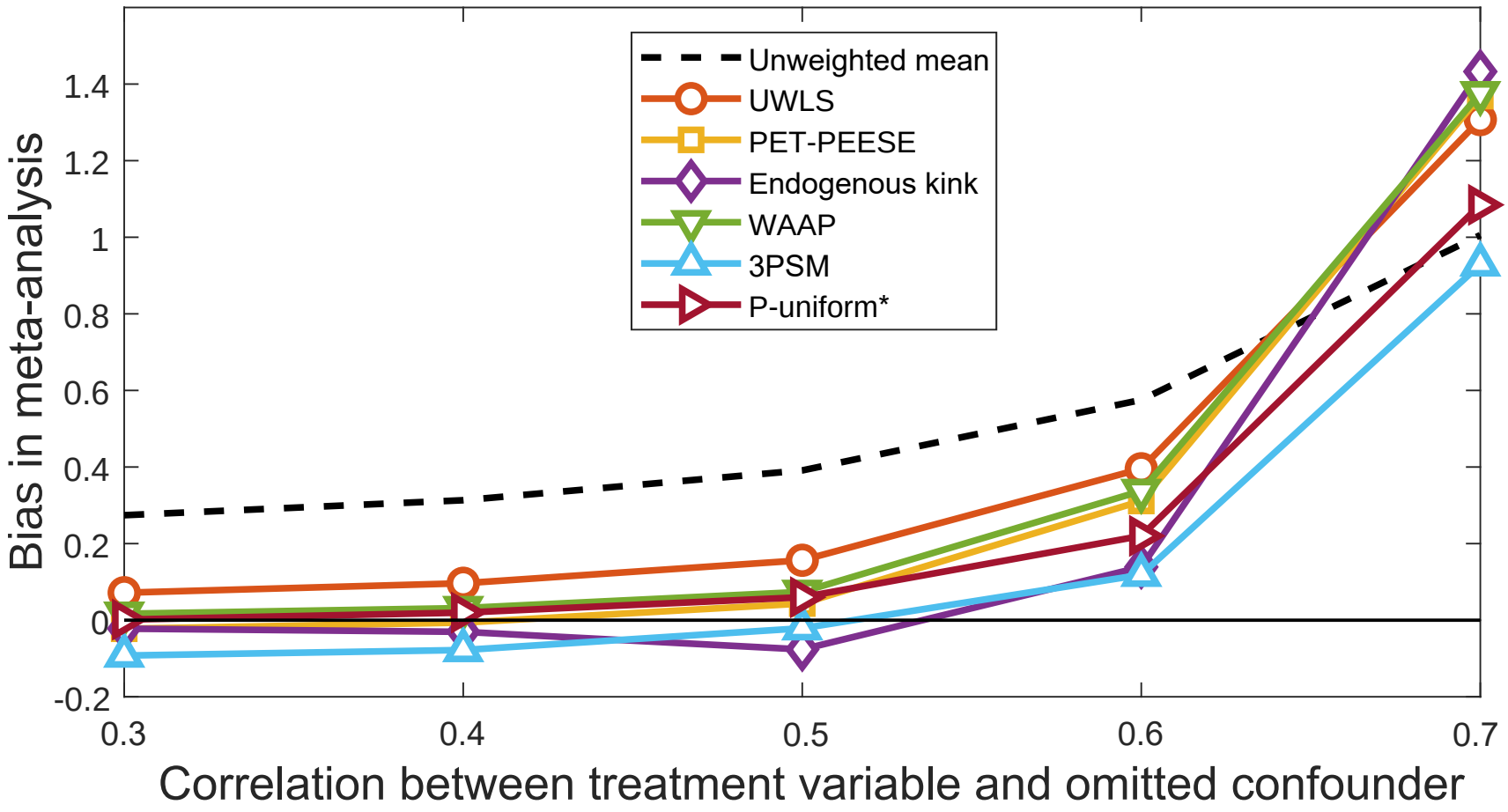
Notes: Adapted from Krueger (1999). Dependent variable: test score percentile.

Standard errors in parentheses.

P-hacking possible on both effect and precision



Standard corrections can fail



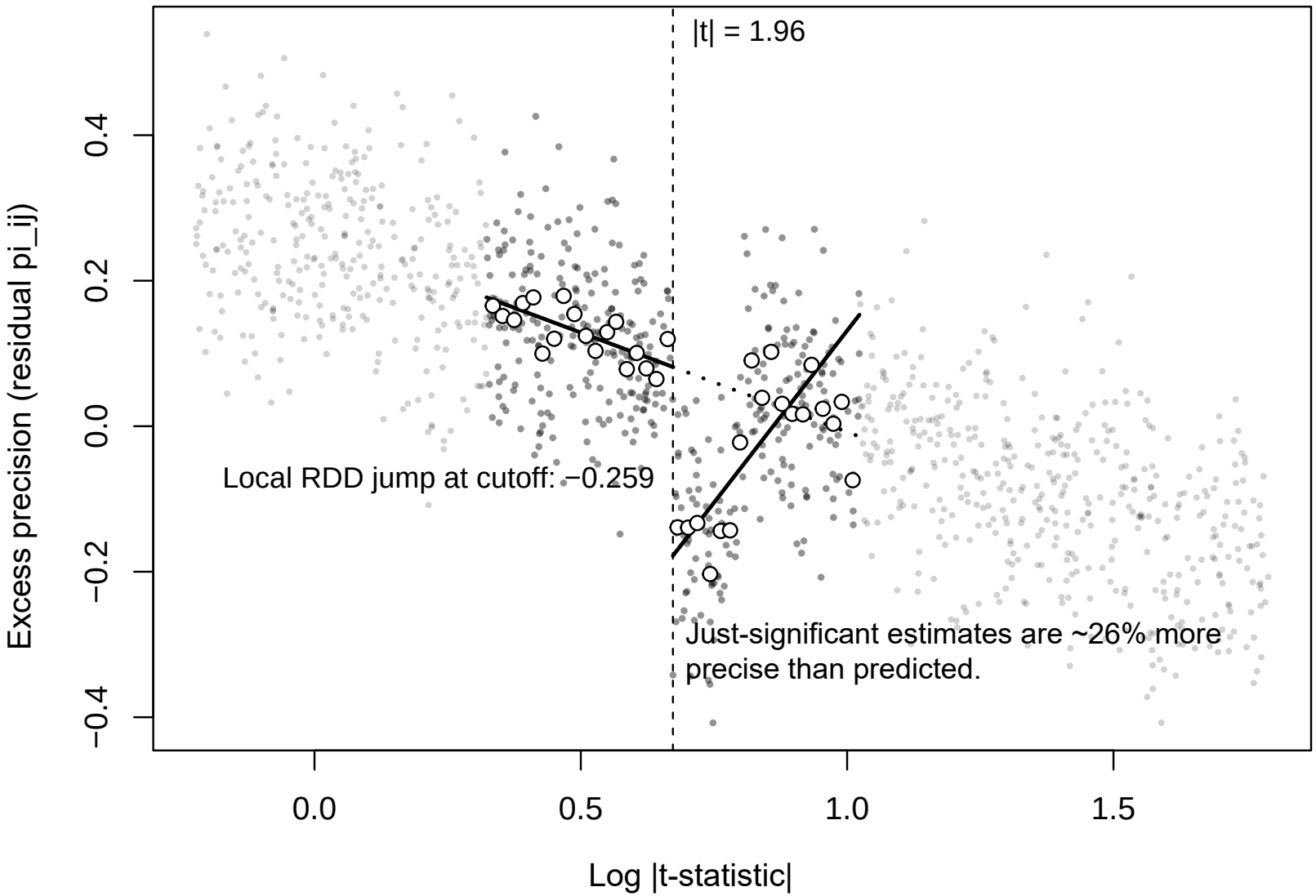
Diagnostic: excess precision

Residual from meta-regression:

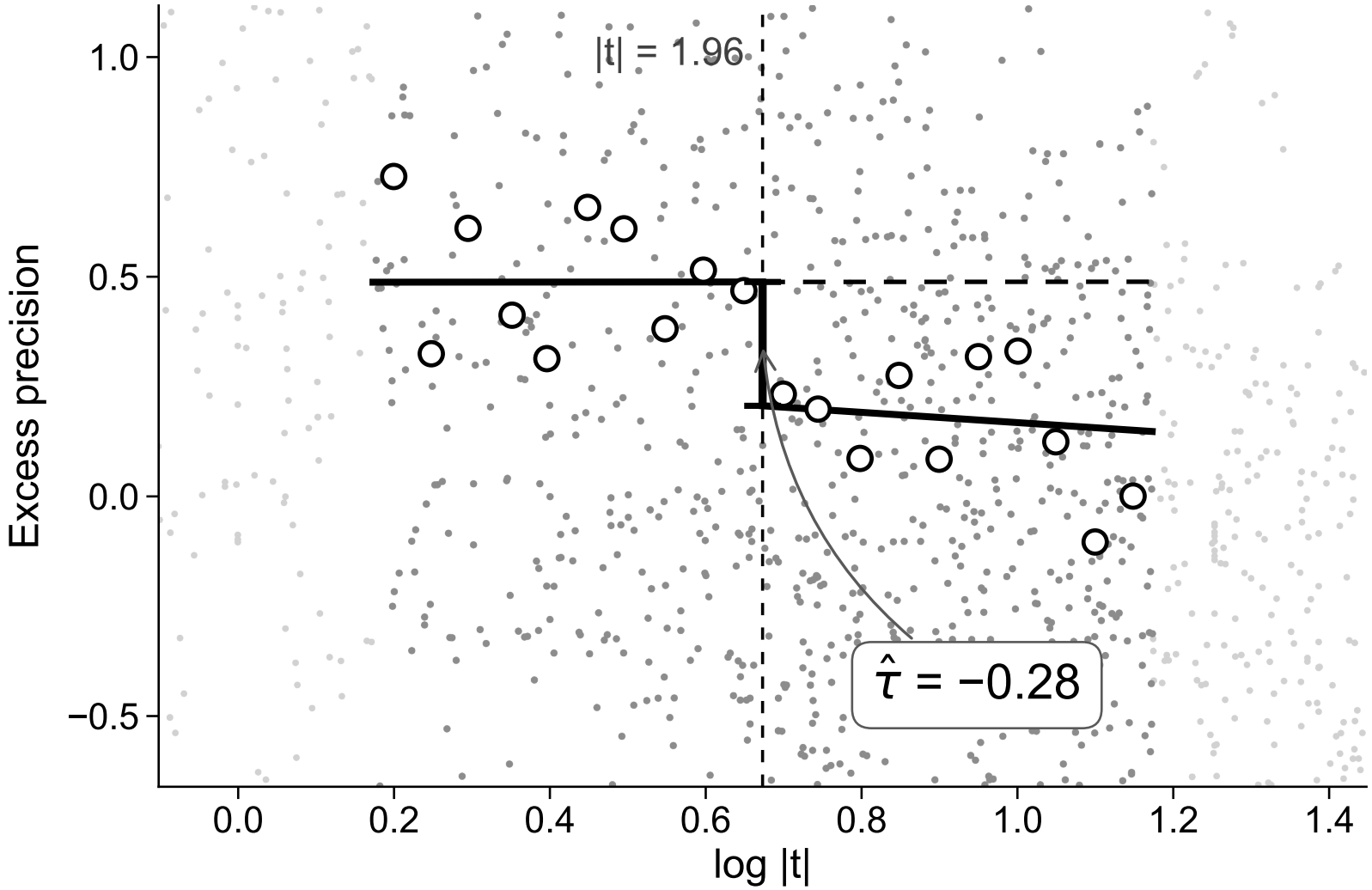
$$\log \mathrm{SE}(\hat{E}_i) = \alpha_0 + \alpha_1 \log N_i + \underbrace{\pi_i}_{<0\text{: excess precision}}$$

- Negative $\pi_i \rightarrow$ reported SE too small relative to $N \rightarrow$ may indicate *p*-hacking.
- Interpretation: $\pi_i = -0.25$ means SE is roughly one quarter smaller than expected.

Residual discontinuity test (simulated)



Pilot: excess sensitivity of consumption



Outside the top-5 journals; 1,039 estimates, 90 studies in the window. Axis trimmed.

Does the picture survive RDD checks?

- **Placebo cutoffs:** fake thresholds at 1.5 and 2.5
  no jump there ($p = 0.73, 0.79$)
- **Donut:** drop the points on 1.96
  $\hat{\tau}$ between -0.30 and -0.23
- **Bandwidth sensitivity:** window $\times 0.5$ to $\times 1.5$
  $\hat{\tau}$ from -0.32 to -0.18
- **Covariate balance:** sample size, publication year
  smooth through the cutoff

Weighted Adjustment Instrumental Variable Estimator

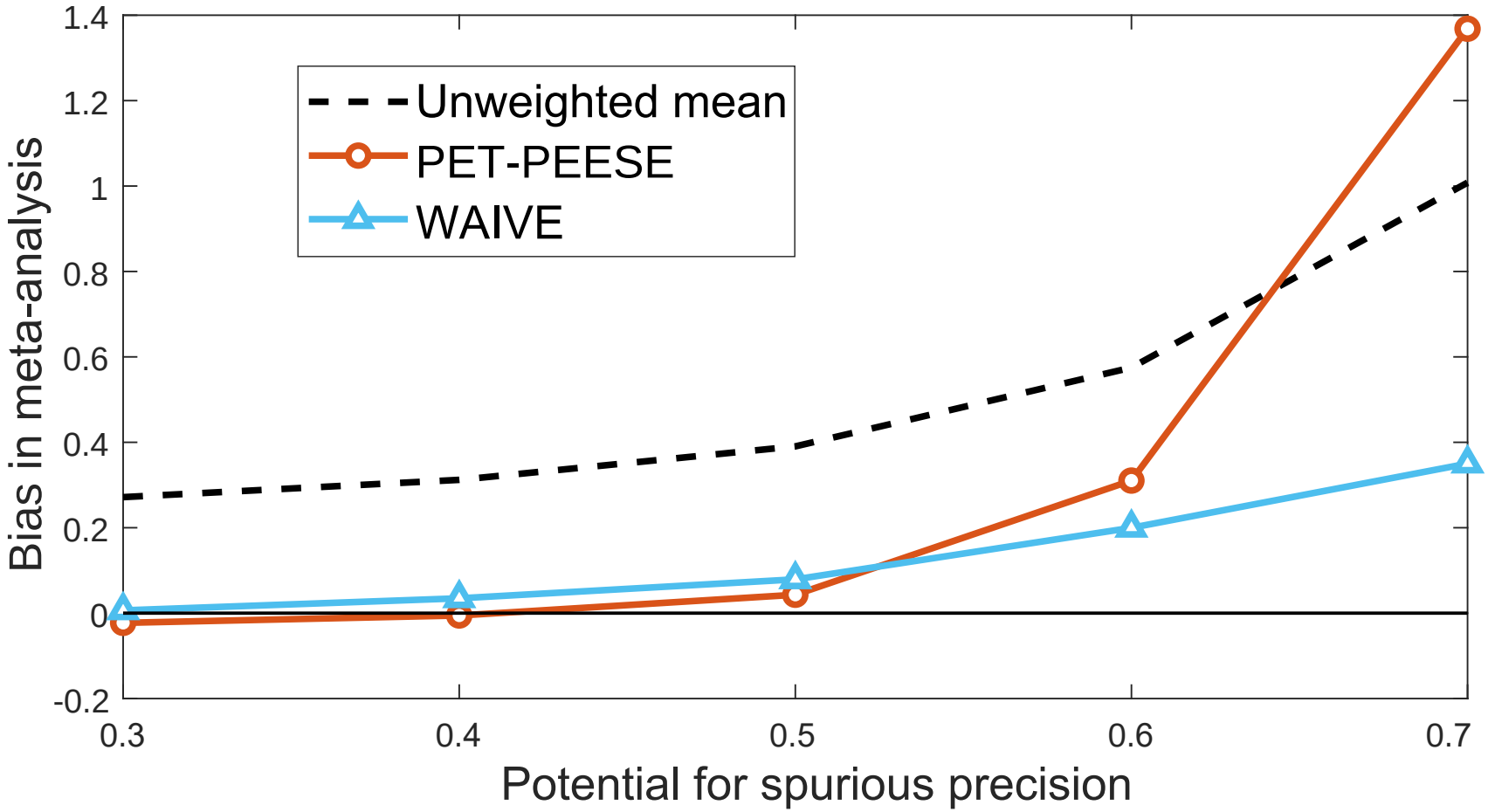
WAIVE combines two adjustments:

1. Replace reported SEs by the component predicted from sample size (instrument). This is MAIVE.
2. Downweight spuriously precise estimates near significance cutoffs. Exponential tilt:

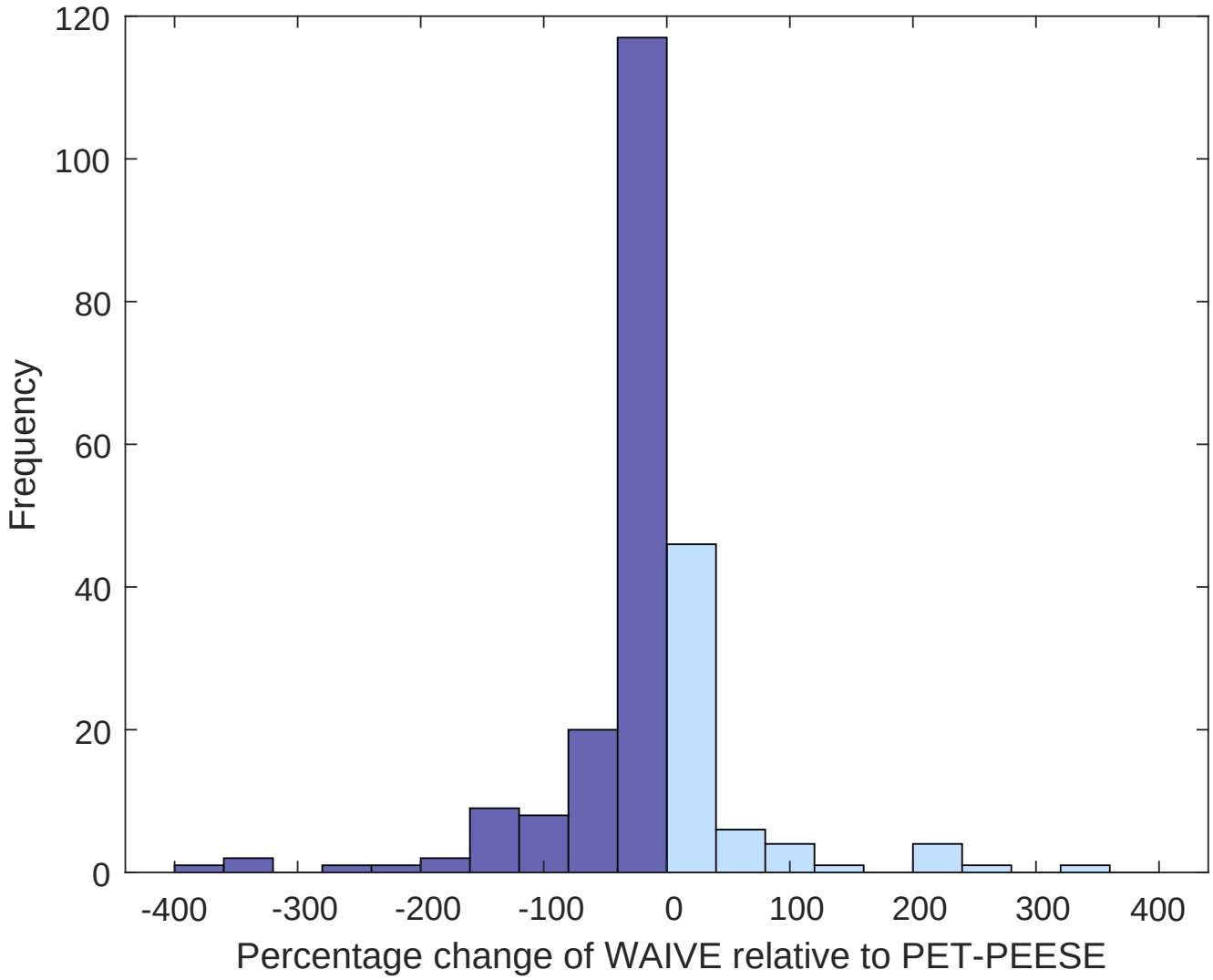
$$\omega_i = \exp[-D_i \max(0, -\pi_i)],$$

- D_i is high when $|t_i|$ is close to 1.96.
- Weight $\omega_i \in (0, 1]$; only excess precision penalized.

WAIVE alleviates the bias in simulations



WAIVE shrinks PET-PEESE in applications



Practical takeaway

The test

RDT: Excess precision should not jump at $|t| = 1.96$.

If it does

MAIVE instruments the standard error with sample size.
WAIVE downweights what is left (more aggressive correction).

Try it

EasyMeta.org · meta-analysis.cz/maive

Papers on instrumental meta

- Irsova Z., P. Bom, T. Havranek, & H. Rachinger (2025): Spurious Precision in Meta-Analysis of Observational Research.
  Nature Communications 16: 8454.
- Irsova Z., T. Havranek, X. Bortnikova, & F. Bartos (2026): Meta-Analysis of Field Studies on Beauty and Professional Success.
  Nature Human Behaviour, R&R.
- Cala P., T. Havranek, Z. Irsova, M. Luskova, J. Matousek & J. Novak (2026): Financial Incentives and Performance: A Meta-Analysis of Experiments in Economics.
  Journal of Political Economy Microeconomics, forthcoming.

Technical appendix

Backup slides for discussion.

Where to look for more

- **Method paper.** Irsova, Bom, Havranek & Rachinger (2025), *Nature Communications* 16:8454.
- **Tools and replication.** easymeta.org · meta-analysis.cz/maive · MAIVE package on CRAN

Excess precision: log first-stage residual

First stage (log form; extension over published MAIVE)

$$\log(\mathrm{SE}_i) \;=\; \alpha + \beta\, \log(N_i) + \gamma' X_i + \pi_i.$$

- $\pi_i < 0 \;\Rightarrow$ reported SE smaller than sample size predicts.
- X_i: pre-specified design features; *not* researcher-chosen inference tweaks.
- Interpretation: $\pi_i = -0.10 \Rightarrow$ SE $\approx$ 10% below prediction (exact: $1 - e^{-0.10} = 9.5\%$).

First stage from MAIVE (Nature Comms 2025); log version used going forward.

Downweighting: D_i on $\log|t|$

Triangular danger-zone kernel (default)

$$D_i \;=\; \max\left\{0,\; 1 - \frac{\left|\log|t_i| - \log(1.96)\right|}{h}\right\}, \qquad t_i = \hat{E}_i / \mathrm{SE}_i.$$

- Compact support: $D_i = 0$ outside bandwidth h; same scale as RDT.

WAIVE weight

$$\omega_i \;=\; \exp[-\,D_i\,\max(0, -\pi_i)] \;\in\; (0, 1].$$

Penalizes only estimates with excess precision *and* $|t|$ near the cutoff.

RDT: residual discontinuity at the significance cutoff

Test of continuity in $E[\pi_i \mid r_i]$ at $r = c$

$$
\begin{aligned}
r_i &= \log|t_i|, \qquad c = \log(1.96), \\
\tau_{\text{RDT}} &= \lim_{r \downarrow c} E[\pi_i \mid r_i = r] - \lim_{r \uparrow c} E[\pi_i \mid r_i = r].
\end{aligned}
$$

- Null: $E[\pi_i \mid r_i]$ continuous at c. Holds if (i) selection works only through the reported t, which reweights the *density* of r and not the conditional mean; and (ii) $E[\pi_i \mid r_i]$ is smooth absent manipulation.
- $\tau_{\text{RDT}} < 0$: just-significant SEs drop below the left-side extrapolation.
- Same scale as π: $\tau = -0.10$ means SEs about a tenth smaller than continuity implies.

RDT: estimation and falsification battery

Estimation

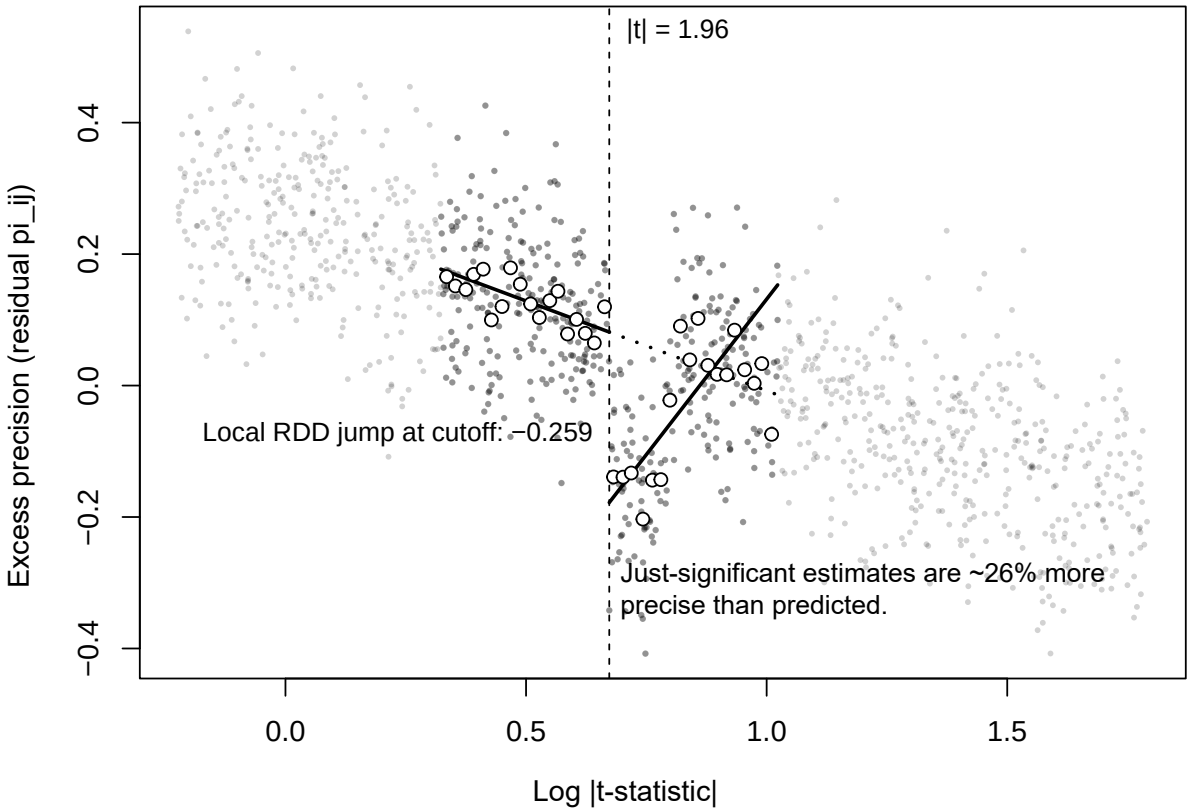
- Outcome: $\hat{\pi}_i$
- Running variable: $\log|t_i|$
- Local linear, both sides
- Triangular kernel
- Bandwidth $h = 0.50$; MSE-optimal as a check
- Cluster bootstrap by study

Falsification battery

- Donut hole around $|t| = 1.96$, sized to the reported digits of SE
- Placebo cutoffs ($|t| = 1.5, 2.5$), one-sided windows
- Predetermined covariates continuous at c
- Bandwidth sensitivity
- Density tests (McCrary; Cattaneo et al.), for context

No global polynomial; identification is intentionally local.

RDT: the picture (simulated)



Simulated: a -0.22 step injected just above the cutoff. The fitted jump reads -0.259 because a straight line through a narrow step overshoots it.

Pilot: sensitivity to the estimator

Same subsample, other estimators

- Cluster bootstrap at $h = 0.50$ (the slide): -0.28, $p = 0.026$.
- `rdrobust`, same h: $+0.05$; MSE-optimal $h = 0.84$: -0.06.

- At the MSE-optimal h the window spans $|t| \approx 1$ to 4.5 and averages it away.

Rounding of SE: the donut has to match the digits

- SE enters both the outcome $(+)$ and the running variable $(-)$, so rounding error δ moves them in opposite directions.
- With ordinary publication selection this lifts $E[\pi \mid r]$ just *below c* more than just above: a negative jump from rounding alone, over a width equal to the rounding scale.
- An SE printed to two decimals as 0.03 carries $\delta \approx 0.1$ in logs.
- A donut of $|t| \in [1.95, 1.97]$ matches rounding in t. Rounding in SE is wider: size the donut from the reported digits, or use a reported t where one exists.

Simulation: data-generating process

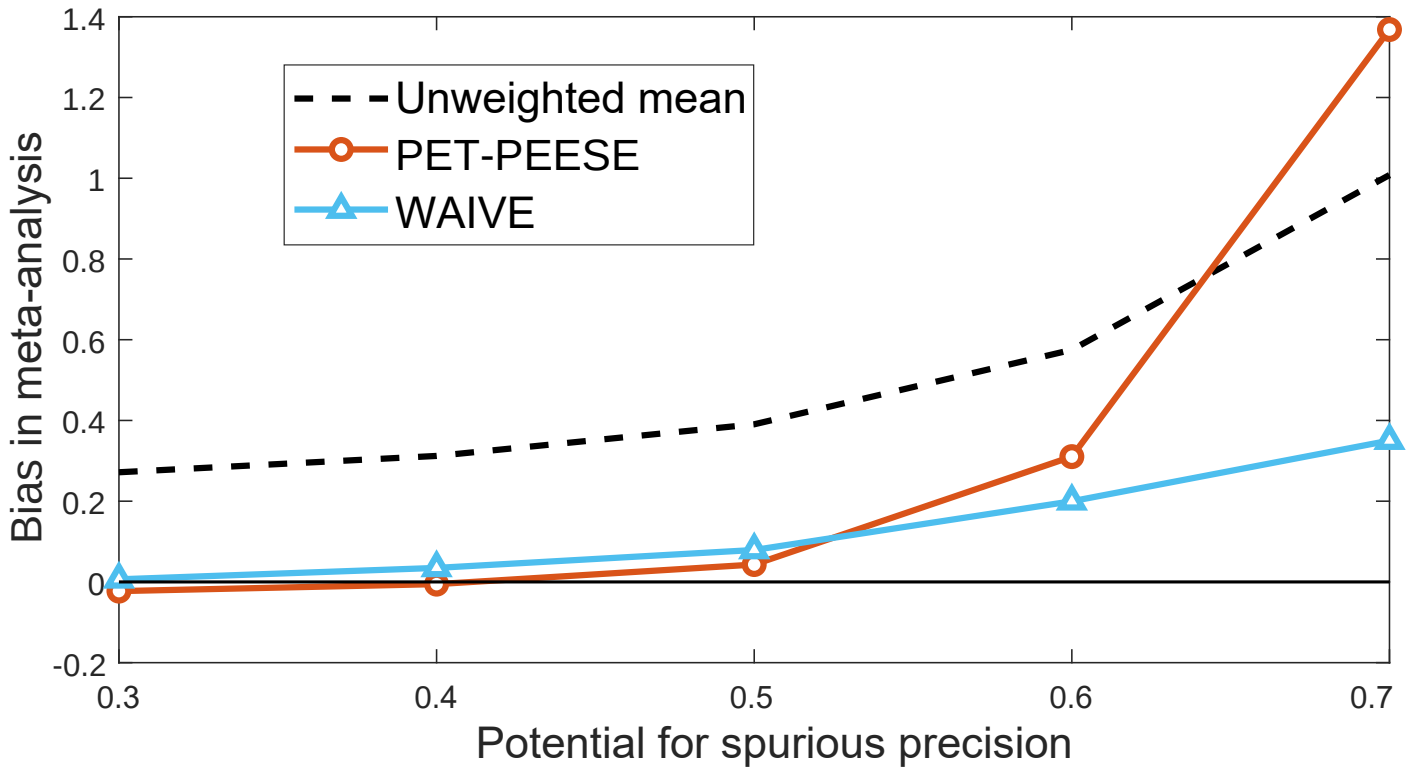
Primary studies

$$Y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + u, \qquad X_2 = \psi X_1 + (1 - \psi)\,\varepsilon.$$

- True effect $\alpha_1 = 1$ (bias on the same scale).
- X_2: true control; replacing it with weaker proxies creates OVB, stronger as ψ rises.
- $M = 80$ studies; sample sizes calibrated to published meta-analyses.
- 50% potential *p*-hackers; search over proxy controls until $t > 1.96$ or $H = 50$ attempts.

Simulation design: MAIVE (Nature Comms 2025); tools at easymeta.org.

Simulation: WAIVE alleviates the bias



At the top of the range WAIVE holds bias to under a third of PET-PEESE.
WAIVE is still experimental; simulation design from MAIVE (2025).